

Nasrallah Rafi

Electrical Engineer — Embedded Systems, IoT & Smart BMS

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Profile

State Engineer in Electrical Engineering and Industrial Systems Control, with a research focus on smart Battery Management Systems, battery analytics, and AI-assisted SOC estimation for LiFePO4 batteries.

Experienced in physical battery modeling, ECM-based simulation, Unscented Kalman Filtering, LSTM residual correction, and digital-twin-based validation for electric mobility applications.

Skilled in MATLAB/Simulink, Python, embedded systems, and industrial data acquisition, with strong interest in predictive modeling, SOH estimation, anomaly detection, and real-time battery health monitoring.

Education

ENSET, Mohammedia – State Engineer's Degree, Electrical Engineering and Industrial Systems Control 2023 – 2026

Faculté des Sciences Ben M'sik, Casablanca – Licence in Physical Sciences (Electronics) 2022 – 2023

Faculté des Sciences Ben M'sik, Casablanca – DEUG in Physics 2020 – 2022

Lycée Technique, Mohammedia – Baccalaureate in Electrical Sciences and Technologies 2019 – 2020

Professional Experience

Emove Véhicules

Engineering Intern (Internship – R&D and Industrialization)

Feb 2026 – May 2026

Project: Systemic resolution of a critical premature shutdown defect (at 20% SoC) and energy gauge instability during abrupt accelerations on electric scooters through a coupled Hardware / Software / Process approach.

- **Digital Twin:** Development of a high-fidelity virtual battery (15S2P configuration, LFP chemistry) integrating dynamic hysteresis models and a digital guard/safety architecture (IEC 62133 / ISO 26262 standards).
- **Estimation Algorithms & AI:** Design and implementation of a hybrid State of Charge (SoC) estimation architecture combining a physical model (ECM 3RC), an Unscented Kalman Filter (UKF), and an LSTM neural network. Development of a residual observer (AI) to compensate for unmodeled phenomena.
- **Advanced Energy Management:** Integration of a bidirectional active balancing system (LTC3300-1) and development of a safety supervisor (Finite State Machine - FSM) for degraded mode management and system protection.
- **Hardware & Embedded Design:** Designed and routed in Altium Designer a 16S Battery Management System (BMS) board based on an STM32 architecture, integrating EMC considerations, power protections, and I2C, SPI, and CAN communication interfaces.
- **Validation & Robustness:** Validation of algorithms via MIL simulation on a digital twin (RMSE < 2%) and experimental validation on an instrumented HiL prototype (Arduino, High-side MOSFET).
- **Quality Management (QMS):** Development and implementation of a strict industrial cell qualification procedure (PR-BAT-01) inspired by the ISO 9001 standard.
- **Supervision and Digitalization:** Creation of a real-time industrial dashboard (MQTT, REST API) for production monitoring, complete traceability of battery packs, and KPI analysis.

SMCV

Mohammedia

Engineering Internship

Jul 2025 – Sep 2025

- Requirement analysis, programming, and supervision of a plastic injection machine (TIA Portal).
- Comprehensive design, study, and automation of a hot nut insertion press (control-command schematics), optimizing productivity and process repeatability.

ONEE

Mohammedia

Technical Internship

Jul 2024 – Aug 2024

- Analysis of Siemens digital relays and development of testing procedures.

Patents

OMPIC n°65133

Automated system and process for welding and assembling corrugated web beam profiles

Issued Nov 2025

Industrial Patent – Lead Inventor.

Technical Skills

Expertise: Modeling & Advanced Control, Embedded Systems & IoT, Industrial Automation & Supervision, Power Electronics & Electrical Grids.

Software: Altium Designer, Siemens TIA Portal, PLCnext Engineer, MATLAB/Simulink, Caneco BT, ETAP, Proteus, Ansys (HFSS/Mechanical), Autodesk Inventor, AutoCAD, Fusion 360.

Programming: C/C++, Python, Ladder/SCL, VHDL.

Languages: Arabic (Native), English (C2 - Advanced), French (B2 - Intermediate).